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REVIEW



The role of microbiota in allogeneic hematopoietic stem cell transplantation

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ABSTRACT

Introduction: Allogeneic hematopoietic stem cell transplantation (Allo-HSCT) is commonly performed to treat a variety of benign and malignant hematological diseases. Acute graft-versus-host disease (GVHD) is a major life-threatening complication that often occurs following allo-HSCT. Recently, improvements in methods to characterize the microbiota have led to a greater appreciation for how frequently and profoundly an alteration in microbial composition, or dysbiosis, can occur in allo-HSCT recipients to better decipher the complex interplay between microbiota and allo-HSCT outcomes.

Areas covered: This article reviews the current knowledge of the microbiota's impact on allo-HSCT outcomes, including effects of microbiota-derived metabolites, and crosstalk between commensals and the allogeneic immune response. This article also summarizes the effects of HSCT and transplant-related procedures on microbiota, and recent developments in interventional strategies.

Expert opinion: A growing body of literature indicates that the composition of the intestinal microbiota can function as a predictive biomarker for the risk and severity of acute GVHD, as well as overall survival, in patients undergoing allo-HSCT. Mechanisms underpinning these associations, however, are not well understood, and clinical strategies that modulate the microbiome to improve outcomes have yet to be fully developed. There is an unmet need to determine mechanisms to improve the efficacy of allo-HSCT.

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Microbiota; dysbiosis; metabolites; allogeneic hematopoietic stem cell transplantation; graft-versus-host disease

1. Introduction

Allogeneic hematopoietic stem cell transplantation (Allo-HSCT) is a potentially curative treatment for both benign hematological conditions and hematologic malignancies [1]. Acute graft-versus-host disease (GVHD) is a post-transplant inflammatory disease that can typically involve the skin, liver, and gastrointestinal tract [2]. Acute GVHD remains a major complication of allo-HSCT, occurring in 20% to 40% of allo-HSCT patients, despite significant improvements in both preventive and clinical management [3].

The human body harbors trillions of resident microbes that can impact on many aspects of health, including playing a role in regulating immune responses [4,5]. A diverse composition of intestinal commensals develops naturally early in life and protects the host from gastrointestinal infections by a variety of mechanisms, including inhibition of domination by pathogens [5,6]. Advances in next-generation sequencing technologies have shed light on previously underappreciated associations between commensal bacteria and risk for diseases. In the last decade, the intestinal microbiota has emerged as an important modulator of allogeneic immune reactions and GVHD-related lethality following allo-HSCT. HSCT and transplant-related procedures such as antibiotic regimens that are commonly used in patients undergoing HSCT can cause drastic changes in the gut microbiota [7]. Disruption of the composition of the microbiome, known as

dysbiosis, is characterized by loss of diversity and reductions of major subgroups of bacterial commensals commonly found in normal individuals [8]. Studies have found that dysbiosis is strongly associated with poor clinical outcomes in allo-HSCT patients [7]. Until recently, however, an association between microbiome injury and major allo-HSCT-related complications was unclear.

Integrative analyses have shown that alterations in microbiome-derived metabolites are associated with development of GVHD [9]. Combined metabolic and bacterial profiling could prove to be a particularly powerful approach for predicting post-transplantation outcomes [10,11]. Understanding the interplay between microbiome-derived metabolites and allogeneic immune response should provide new insights into developing microbiome-based prevention and interventional strategies. Here, we discuss the emerging concepts arising from recent investigations into how the intestinal microbiome affects clinical outcomes in patients receiving allo-HSCT.

2. Loss of microbiota diversity and survival outcomes

Obligately anaerobic commensal bacteria that reside in the intestine have been found to play a critical role in maintaining normal gut homeostasis [12]. Dysbiosis in allo-HSCT patients commonly manifests with loss of obligately anaerobic commensal bacteria,

Article highlights

- Intestinal microbiota has emerged as a novel therapeutic target to potentially improve clinical outcomes of allo-HSCT recipients.
- Changes in metabolomic profiles have been associated with development of acute GVHD in recipients of allo-HSCT.
- Many studies have identified associations between high diversity of the intestinal microbiota and a lower risk of transplant-related mortality.
- Commensal bacteria can modulate immune responses and promote effector T cell responses.
- Strategies to manipulation the microbiome such as FMT have been shown to directly change the recipient's gut microbiota composition and may mediate clinical benefits.

This box summarizes key points contained in the article.

expansion of pathogenic bacteria, and a reduction in overall microbiome diversity [13,14]. Dysbiosis can contribute to intestinal disease pathogenesis by producing increased gut permeability and disrupting normal barrier function, in turn leading to further gastrointestinal damage. Traditionally evaluated with microbiological culturing techniques, dysbiosis can now be identified using high-throughput 16S rRNA gene amplicon-based next-generation sequencing [7].

Quantitative species-based measures such as inverse Simpson index and Shannon diversity index were often used to summarize and compare the microbiome alpha diversity in different communities [15]. To examine the potential impact of the intestinal microbiota on allo-HSCT outcomes, Taur et al studied allo-HSCT recipients from a single institution, examining fecal samples collected at the time of neutrophil recovery [16] and found that higher intestinal microbiome diversity was significantly associated with improved overall survival and reduced transplant-related mortality compared to patients with lower diversity (Inverse Simpson less than 2). Another study demonstrated that an increase in intestinal bacterial diversity was in particular associated with reduced mortality from GVHD, while associations between diversity and malignant relapse were not observed [17]. Importantly, the finding of an association between high diversity of the intestinal microbiota and a lower risk of transplant-related mortality was later confirmed in a study with large multi-center international cohorts [18].

3. Microbiota changes, allo-HSCT and related procedures, and complications

3.1. Microbiota and host physiology

Host-microbe interactions occur at mucosal surfaces, and the microbial communities at different human body sites are known to be compositionally distinct [19]. One of the largest is the human intestinal mucosa, estimated to be between 250 and 400 m², home to between 10¹³ to 10¹⁴ microbes [4,20]. The intestinal epithelium is a single layer of cells comprised of intestinal epithelial cells (IECs), which includes specialized cells such as Paneth cells, goblet cells, and intestinal stem cells. IECs provide two major functions: (1) serving as a protective barrier separating the intestinal lining from the luminal components;

and (2) absorbing nutrients. Beneficial commensal intestinal bacteria further supplement the intestinal epithelial barrier by protecting the host from colonization by pathogens [21].

Paneth cells are a specialized subset of epithelial cells residing in crypts, and are largely found in the small intestine. They produce secretory antimicrobial molecules such as α -defensins and play an essential role in regulating the microbiota community [22]. Loss of Paneth cells during GVHD causes a reduction of α -defensins, which is accompanied by changes in the diversity and the composition of intestinal microbial communities [23,24].

3.2. Antibiotic use and GVHD

In 20–60% of HSCT patients, a complication of the HSCT process is developing a systemic infection [25]. Prophylactic antibiotic regimens and systemic administration of broad-spectrum antibiotics are commonly used in allo-HSCT to prevent and treat bacterial infections [26]. The relationship between antibiotic treatment and transplant-related complications has been investigated by several groups. In one study, clinical use of imipenem-cilastatin or piperacillin-tazobactam, which are both broadly active against many bacteria, was associated with an increased incidence of lower gastrointestinal GVHD, while there was no impact seen in patients treated with more narrow-spectrum antibiotics including aztreonam or cefepime [27]. Interestingly, allo-HSCT mouse recipients treated with imipenem-cilastatin showed an expansion of *Akkermansia muciniphila* and reduced colonization by *Clostridiales* [27]. These findings suggested that commensal bacteria targeted by certain antibiotics and spared by others may protect against subsequent GVHD. Farowski et al [28] similarly found that the use of penicillin derivatives and carbapenems was associated with a higher incidence of acute GVHD. Another retrospective study in Japan found that long use of carbapenem, but not penicillin, for greater than 7 days may be a risk for intestinal GVHD [29]. The timing of antibiotic administration can also affect clinical outcomes. In a study of two institutional cohorts, early administration of systemic antibiotics before hematopoietic stem cell infusion (before day 0) was significantly associated with a high transplant-related mortality compared to those who did not receive antibiotic treatment or experienced late administration of antibiotics after hematopoietic stem cell infusion (on day 0 or thereafter) [30]. Together, these results indicate that damage of the microbiome, particularly by broad-spectrum antibiotic regimens, may have a detrimental effect on outcomes.

Several bacteria in the intestinal tract have been reported to be associated with clinical outcomes in allo-HSCT. Higher abundances of bacteria belonging to the genus *Blautia* was associated with improved overall survival [17], and loss of *Blautia* was associated with antibiotic use and longer duration of total parenteral nutrition. Loss of commensal bacteria such as *Blautia* is often accompanied by expansions of potentially pathogenic bacteria. Eriguchi et al found that development of GVHD in mice is associated with an outgrowth of *Escherichia coli*, and oral administration of polymyxin B reduced *E. coli* levels and ameliorated GVHD [24]. Another pathogen of

intestinal origin, *Enterococcus*, was recently found to be associated with poor outcomes, including significantly reduced overall survival and development of acute GVHD. Murine modeling experiments uncovered a mechanism for outgrowth of *Enterococcus* that depended on utilization of lactose [31].

Early studies by Bekkum et al [32] and Jones et al [33] in mice demonstrated that the intestinal microbiome contributes to acute GVHD. Thus, early clinical studies incorporated prophylactic broad-spectrum oral antibiotics, with a goal of attempting to achieve near-total bacterial decontamination. Results of a later prospective study by Beelen et al indicated that anaerobic intestinal bacteria could especially contribute to GVHD, when they found that bone marrow transplant recipients treated with both metronidazole and ciprofloxacin as bacterial decontamination medication exhibited lower GVHD severity compared to ciprofloxacin alone [34].

More recently, however, Holler et al reported that use of this same prophylactic regimen was associated with loss of diversity in the gut and expansion of *Enterococcus* [7], and an institutional practice change to rifaximin resulted in less microbiome injury and substantial reductions in acute GVHD and transplant-related mortality [35]. And, as described earlier in this review, use of broad-spectrum antibiotics to treat neutropenic fever has been associated with increased acute intestinal GVHD across several centers. In contrast, a recent retrospective study found that oral vancomycin prophylaxis, which is very disruptive to the microbiome [13], effectively prevented CDI in recipients who had undergone allo-HSCT but was not associated with an increased risk of GVHD [36]. One likely explanation for these seemingly contradictory reports is that prophylactic antibiotics may be producing variable effects in patients, perhaps in relation to different geographic locations. A retrospective evaluation in a cohort of children undergoing allo-

HSCT, all treated with oral decontaminating antibiotics, found that microbiological evidence of decontamination was only observed in half of the patients, and successful decontamination was strongly associated with reduced acute GVHD [37], compared to higher rates of GVHD in children unsuccessfully decontaminated. Thus, developing a one-size-fits all antibiotic strategy to reduce acute GVHD may not be possible and a more customized approach based on an individual patient's intestinal microbiome may be more successful. We expect that future clinical studies will identify additional bacterial subsets associated with either protection or harm. These studies, as well as experimentation that mechanistically explains these associations, will help improve our understanding of why these findings that link microbiome disruption to poor allo-HSCT outcomes are reproducibly and consistently seen across institutions and geography.

3.3. Conditioning therapy and GVHD

The allo-HSCT procedure consists of a conditioning regimen, which includes chemotherapy, radiation therapy or antibody therapy to deplete both malignant and immune cells in patients, followed by infusion of donor hematopoietic cells [38]. A side-effect of high-dose chemotherapy conditioning is intestinal injury, which can be accompanied by shifts in microbiota composition

[39]. Jenq et al [17] have examined the association between *Blautia* and GVHD outcomes in patients divided into subgroups by the intensity of conditioning regimens: standard intensity myeloablative, reduced intensity myeloablative, and nonmyeloablative. *Blautia* abundance was significantly associated with reduced GVHD in patients receiving nonmyeloablative conditioning. However, this association was not statistically significant in patients who received reduced intensity myeloablative and myeloablative conditioning [17]. One study found that conditioning resulted in a reduction in alpha diversity, with significant decreases in *Firmicutes* and increases in *Escherichia* species. This shift in intestinal microbiome and a decrease in microbial diversity may induce more complications and contribute to the development of intestinal GVHD [40].

Patients are at much higher risk for developing a systemic infection when they are neutropenic, most commonly during a period of 10–12 days after conditioning therapy [41]. Bacteremia is a particularly serious complication and requires immediate antibiotic therapy. An examination of the microbiome of allo-HSCT patients with bacteremia found that reduced diversity of intestinal flora was frequently observed, along with domination by bacteria that were similar to those found in the bloodstream [42]. Prophylactic antibiotics during the neutropenic period are recommended to prevent bacterial infections [26]; however, the optimal choice of prophylactic antibiotics in the allo-HSCT population remains unclear. More studies are needed to develop the strategy of antibiotic prophylaxis to prevent GVHD and pathogenic infections.

3.4. Pulmonary complications

Patients undergoing HSCT often experience pulmonary complications, which contribute to morbidity and mortality after HSCT [43,44]. Harris et al studied the association between gut microbiota and pulmonary complications in recipients who underwent allo-HSCT and found that low baseline diversity and early domination of γ -proteobacteria were associated with an increase in pulmonary complications [43]. These findings suggested that systemic translocation of bacteria or bacterial products from gastrointestinal tract to the lungs, or alternatively, indirect lung injury, might be leading to subsequent pulmonary complications.

4. Microbiota influence immune response

Commensal bacteria can play an influential role in modulating immune responses locally and systemically. The mechanisms underlying the microbiota's effects in orchestrating immune responses and modulating allo-HSCT outcomes are not entirely understood. Ingham et al. assessed the interplay between immune reconstitution, clinical outcomes, and gut microbiota in pediatric recipients of allogeneic HSCT [45] and observed that patients dominated by higher abundances of Ruminococcaceae and Lachnospiraceae showed a more rapid reconstitution of nature killer (NK) cells and B cells as well as no or mild acute GVHD, and improved overall survival [45].

Interestingly, certain commensal bacteria can function to promote effector T cell responses. In a study looking for associations between the composition of intestinal bacterial and

response of hematological malignancy to allo-HSCT, higher abundances of *Eubacterium limosum* were found to be associated with a decreased risk of relapse and delayed progression of disease [46]. Another study in mice found that an 11-strain consortium of bacterial strains was highly effective in inducing IFN γ + CD8 T cells and augmented anti-tumor immunity; interestingly, this consortium included a member identified as *Eubacterium limosum* [47]. The behavior of CD4 + T helper cell populations can also be influenced by intestinal commensal bacteria. Colonization with segmented filamentous bacterium in mice was demonstrated to be sufficient to induce naïve intestinal CD4 + T cells to differentiate into T helper 17 (Th17) cells that produce IL-17 and IL-22 [48]. In healthy human volunteers, circulating memory CD4 + T cells can be identified that mount responses to intestinal bacteria, producing both Th17 cytokines and T helper 1 (Th1) cytokines [49]. In inflammatory diseases, the dynamic balance between Th17 cells and regulatory T cells (Tregs) may be particularly important in disease pathogenesis, including in the setting of GVHD [50,51]. Interestingly, interleukin-6 (IL-6) is a key inflammatory cytokine that functions to induce effector Th17 differentiation, altering the balance between Th17 cells and Tregs, and blockade of IL-6 signaling with anti-IL-6 R antibody treatment has been recently shown to significantly attenuate GVHD severity [52].

5. Microbiota-derived metabolites modulate GVHD

Metabolites produced by the intestinal microbiome during normal consumption of substrates derived from food or host-derived glycans can have surprisingly potent effects on the host. Increasingly, these metabolic byproducts are being found to be critically important for maintaining normal homeostasis. Disruption of the gut microbiota leads to major changes in metabolomic profiles, which in turn have been associated with development of acute GVHD in recipients of allo-HSCT. (Table 1) [9,53] Among the most abundant intestinal microbiome-derived metabolites are short-chain fatty acids (SCFAs), which include butyrate, acetate, and propionate. Butyrate has been identified as particularly impactful on the intestinal tract. It acts as a primary source of energy for intestinal epithelial cells, and also helps to regulate intestinal immunity. Many butyrate-producers, such as *Roseburia intestinalis* and *Eubacterium rectale*, tend to be found in higher abundances adjacent to intestinal mucin where they contribute to intestinal homeostasis [54,55]. Intragastric administration of butyrate in murine models of GVHD led to a significant survival benefit, improving tight junction integrity. A similar benefit in reducing GVHD was seen with introduction of a consortium of *Clostridia* strains which together increased butyrate levels [56,57]. One pathway by which SCFAs can mediate effects on the host is through the nutrient-sensing G-protein coupled receptor 43 (GPR43), which is expressed on a variety of cells including IECs. GPR43 was recently shown to be important for mediating the benefit of butyrate in reducing GVHD by activating NLRP3 inflammasome via ERK phosphorylation [56]. Surprisingly, however, researchers have found that once damage occurs, the presence of butyrate can in turn hinder recovery, as the presence

Table 1. Microbial metabolites in modulating GVHD.

Microbiome-derived metabolites	Immune response	GVHD related outcomes
Short-chain fatty acids (SCFAs): acetate, butyrate, and propionate	Regulate intestinal immunity.	Administration of butyrate in murine models of GVHD led to a significant survival benefit.
Trimethylamine N-Oxide (TMAO)	Induce M1 macrophage polarization and subsequently activate Th1 and Th17 differentiation and proliferation.	Administration of TMAO in mice led to aggravate GVHD-related lethality.
Tryptophan-derived metabolites	Many tryptophan-derived have been reported as ligands for the aryl hydrocarbon receptor (AhR).	Lower urinary 3-indoxyl sulfate (3-IS) detected in allo-HSCT recipients within the first 10 days after allo-HSCT was associated with significantly higher transplant-related mortality, mostly as a consequence of GI GVHD.

of butyrogenic bacteria was found to be associated with developing treatment-refractory GVHD [58].

Metabolic byproducts produced by commensal bacteria can mediate substantial physiological impacts on the behavior of immune cell populations as well as epithelial cells [59]. Tregs play a critical role in mediating anti-inflammatory responses in the body, including in the intestinal tract in the setting of GVHD. Butyrate and propionate can directly induce Treg function as well as facilitate extrathymic differentiation of Tregs [60]. Interestingly, subsets of bacteria that do not produce butyrate can nevertheless impact on butyrate concentrations. The probiotic *Lactobacillus rhamnosus* GG, for example, was shown to have an indirect role supporting the development of Treg cells via enriching butyrate-producing bacterial strains [61]. Another important microbial metabolite, choline-generated trimethylamine N-oxide (TMAO), has been shown to induce M1 macrophage polarization and subsequently activated Th1 and Th17 differentiation and proliferation, which led to aggravated GVHD-related lethality in mice [62].

Indole is a metabolite produced by bacteria during degradation of tryptophan. Intestinal bacteria that express tryptophanase can catabolize tryptophan to produce indole and indole derivatives [63], which include indole-3-carboxaldehyde (ICA), was shown to reduce damage to IECs damage and reduced subsequent GVHD in mice after allo-HSCT [64]. Indole is metabolized by the liver into 3-indoxyl sulfate (3-IS), and is ultimately excreted in the urine in humans [10]. In a recent study, lower 3-IS levels detected in the urine within the first 10 days after allo-HSCT were associated with significantly higher transplant-related mortality and poor overall survival due to an increase in intestinal GVHD [10]. Interestingly, this study found that a higher abundances of *Lachnospiraceae* and *Ruminococcaceae*, which are members of the order of *Clostridiales*, was associated with higher urinary 3-IS levels; thus, lower urinary levels of 3-IS indicate that there may be reductions in *Clostridiales* within the microbiome [10]. Michonneau et al [9] reported that additional tryptophan-

derived metabolites (indoleacetate, indoleacetylglutamine, and indolepropionate) were depleted in patients with active acute GVHD. Notably, these compounds are known to act as ligands for the aryl hydrocarbon receptor (AhR) and can regulate allogeneic T cell reactivity, and thus low levels seen during acute GVHD could contribute to GVHD physiology [9]. Together, these studies demonstrate that a variety of microbial-derived metabolites are capable of modulating acute GVHD severity in mouse models and may be potential targets for preventing or treating acute GVHD in patients.

6. Manipulation of microbiome in allo-HSCT

Given the body of work summarized above, the intestinal microbiota has emerged as a novel therapeutic target to potentially improve clinical outcomes of allo-HSCT recipients. Major categories of microbiome-focused strategies include nutritional support, probiotics, prebiotics, and fecal microbiota transplantation (FMT), all of which aim to supplement or modulate the biology of the microbiome [65].

6.1. Nutritional support

Reduction in oral nutrition is common in patients undergoing allo-HSCT, due to nausea, anorexia, and mucositis side-effects from conditioning therapy. Parenteral nutrition (PN) is commonly administered through processes that vary widely across centers that perform HSCT. This approach, however, can be associated with complications such as infections and dysbiosis of gut microbiota. Seguy et al showed that enteral nutrition (EN) after myeloablative conditioning may improve posttransplant outcomes by reducing mortality and aGVHD incidence [66]. Many studies have explored the impact of EN on gut microbiota composition, including supporting the recovery of a diverse eubiotic gut microbiota. In a cohort of 20 pediatric patients who underwent HSCT, patients receiving PN had a reduction in SCFA levels following HSCT compared to pre-HSCT. In contrast, patients receiving EN post-transplant had a swift recovery of the gut microbiome with higher levels of SCFAs, including butyrate, acetate, and propionate [67]. Similar findings were observed in an adult cohort of 23 patients undergoing HSCT, where those who received EN had greater abundances of SCFA-producing taxa such as *Ruminococcus bromii* and *Faecalibacterium prausnitzii* compared to patients receiving predominantly PN [68]. Incidence of blood stream infections (BSI) may also be affected by the type of nutritional support. Zama et al reported that a significant reduction in the incidence of BSI was observed in allo-HSCT recipients receiving EN [69]. Although PN has been widely preferred after allo-HSCT, increasing evidence supports repleting nutrition using the enteral route. Additional multicenter randomized controlled studies are needed in the allo-HSCT setting to better determine the optimal approach to administer nutrition.

6.2. Probiotics

Probiotics can be either a single strain of bacteria or mixture of strains that are produced and orally consumed, with a goal of

improving intestinal homeostasis. There is some evidence that probiotic-based therapies can modulate the intestinal microbiome and mediate anti-inflammatory effects in mice. In allo-HSCT, Gerbitz et al reported that mice receiving *Lactobacillus rhamnosus* GG had reduced inflammation and bacterial translocation, which led to less acute GVHD and improved survival [70]. A randomized trial evaluating supplementation with *Lactobacillus rhamnosus* GG in allo-HSCT recipients, however, did not appear to appreciably modify the microbiome or reduce GVHD [71]. In a pilot trial, the administration of another probiotic, *Lactobacillus plantarum*, was safe and reasonably tolerated in children and adolescents undergoing allo-HSCT [72]. Currently, a randomized phase III trial aiming to evaluate the efficacy of orally administered *Lactobacillus plantarum* in preventing the development of GI GVHD is underway (NCT03057054).

6.3. Prebiotics

Prebiotics are non-digestible food ingredients that can be fermented by bacteria in the distal small intestine and colon, and include dietary fiber, resistant starches, as well as other saccharides [73]. Prescribing dietary fiber and other prebiotics has been studied as a means of promoting gut health [74,75], and can boost levels of SCFAs [76], which in turn can support tissue repair and regulate inflammation during gut injury [56,57], and preserve epithelial integrity in allo-HSCT [76,77]. In clinical trials, potato starch (NCT02763033) and supplemental fructose oligosaccharides (NCT02805075) are currently being explored as beneficial prebiotics in allo-HSCT patients. One challenge will be heterogeneity in the response of patients to different prebiotics, depending on their microbiome composition and enzymatic capabilities.

6.4. Fecal microbiota transplantation (FMT)

Outside of allo-HSCT, several randomized clinical trials have demonstrated a clear benefit from reconstituting the intestinal microbiota using FMT. Clinical settings with the most established supportive evidence include recurrent *Clostridium difficile*-associated infections and inflammatory bowel disease [78,79]. FMT can be provided by a known acquaintance of the patient, including family or household members, or alternatively from a healthy donor [80]. FMT has been studied in allo-HSCT patients primarily in four clinical settings (Table 2): 1) treatment of recurrent *Clostridium difficile*-associated infection [81], 2) treatment of steroid-refractory or steroid-dependent acute gut GVHD [82–85], 3) prophylaxis in patients with microbiome injury sustained early during the allo-HSCT process, and 4) treatment of intestinal colonization by multi-drug-resistant bacteria (MDRB) [86,87]. We will focus here on the GVHD and prophylactic studies. In a pilot study, Kakihana et al reported that a case series of allo-HSCT patients with steroid-resistant or steroid-dependent intestinal GVHD, responded to FMT entirely or partially with no adverse events attributed to FMT [85]. A larger study of 15 patients was recently reported, with 10 patients responding within one month after FMT [88]. Together these studies indicate that FMT has clinical promise and larger, randomized prospective

Table 2. Summary of recent FMT clinical studies in allo-HSCT patients.

Clinical setting	Sample size	FMT route	Findings	Reference
Treatment of recurrent <i>Clostridium difficile</i> -associated (CDI) infection	7 patients underwent HSCT	Naso-jéjunal (N-J) administration or colonoscopy	FMT was safe and effective therapy for CDI in all patients and may confer benefits.	Webb et al., 2016 [81]
Treatment of steroid-refractory or steroid-dependent acute gut GVHD	4 patients	Nasoduodenal tube	All patients responded to FMT (3 complete responses and 1 partial response).	Kakihana et al., 2016 [85]
Treatment of steroid-refractory following allo-HSCT	3 patients	Colonoscopy	All three patients responded clinically to FMT.	Spindelboeck et al., 2017 [84]
Treatment of steroid-refractory Patients diagnosed with steroid refractory GI tract GVHD	1 patient underwent allo-HSCT 8 patients	Capsules Nasoduodenal tube	Oral FMT for steroid-refractory acute gut GVHD was feasible. All patients achieved progression-free survival. No severe adverse events were observed during and after FMT.	Kaito et al., 2018 [82] Qi et al., 2018 [83]
Prophylaxis in patients with microbiome injury sustained early during the allo-HSCT process	13 patients	Capsules	FMT was associated with expansion of microbiome diversity in transplant recipients.	DeFilipp et al., 2018 [89]
Restoration of gut microbiota diversity and microbial composition to the pre allo-HSCT state	Randomized 25 allo-HSCT patients (14 received auto-FMT; 11 did not)	Rectal administration	FMT increased microbial diversity.	Taur et al., 2018 [90]
Treatment of intestinal colonization by multidrug-resistant bacteria	10 patients	Either by enema or via nasogastric tube	FMT was safe for all patients, except one patient had constipation.	Battipaglia et al., 2019 [86]
Treatment of intestinal colonization by multidrug-resistant bacteria	2 patients underwent allo-HSCT	Intranasal probe	One patient achieved complete remission without complications, whereas therapeutic efficacy was not achieved in the second patient. FMT eliminated drug-resistant pathogens, but not multidrug resistant bacteria.	Biernat et al., 2020 [87]
Patient diagnosed with symptoms of steroid refractory or steroid-dependent, acute or late-onset acute intestinal GVHD	17 allo-HSCT patients (15 evaluable patients; 2 received other GVHD medication were not included)	Nasoduodenal tube	Donor FMT was tolerated by all patients. 10 of 15 patients achieved complete response within one month after FMT, accompanying by an increase in gut microbial diversity. Complete response to donor FMT was associated with overall better survival at 24 weeks after donor FMT.	Van Lier et al., 2020 [88]

studies are warranted. Steroid-dependent patients also seem to be more likely to respond to FMT than steroid-refractory patients with severe involvement, which could guide the design of such a prospective study.

In the prophylactic setting, FMT has yet to be shown to mediate a clinical benefit, but appears to be safe and well tolerated, and can help address microbiome injury sustained during the allo-HSCT process. In one pilot study, 13 allo-HSCT patients received FMT in capsule form from a third party [89]. In another randomized study of 25 allo-HSCT patients, autologous fecal samples were banked from individual patients, and then in some patients re-introduced following neutrophil recovery after allo-HSCT [90]. Both studies indicated that the procedure was generally well tolerated, with clear evidence of improvements in patient microbiome profiles. Larger studies will be required to evaluate for a potential clinical benefit of prophylactic FMT.

The increasing rise of MDRB is a clinical challenge, especially for allo-HSCT recipients. Battipaglia et al reported that FMT is safe to treat patients carrying MDRB; no other major adverse events were observed except one patient developed constipation and two patients experienced grade I diarrhea after FMT [86]. Seven of ten patients undergoing allo-HSCT were successfully decolonized with a median follow up of 13 months (range from 4 to 40 months) [86]. Biernat et al presented two cases of allo-HSCT patients who underwent FMT to treat MDRB colonization, one achieved decolonization without complications, whereas therapeutic efficacy was not achieved in the second patient [87].

There are many aspects of the FMT procedure that have yet to be optimized. Decisions regarding route of delivery (enema, capsules [89], nasoduodenal infusion [82]), timing and frequency, and donor source (autologous, 3rd party single donor, 3rd party pooled donor, selected 3rd party donors) are all factors that have to yet be determined in clinical studies. Careful screening of FMT donors will also be essential to maximally prevent the chances for potential harm, including eliminating the possibility of introducing antibiotic-resistant bacteria in the FMT product, which has been reported to lead to bloodstream infection with FMT-derived bacteria and one case of mortality [91].

7. Beyond the lower intestinal microbiome

The oral cavity is the entry point to the digestive tract, and associations between the composition of the oral microbiome and pathophysiology of certain diseases have been observed previously [92,93]. The HSCT process has been found to result in distinct changes to oral and salivary microbial composition. In comparison to the gut microbiome, associations between the oral microbiome and outcomes are not as well studied but is an active area of research. One study examined the oral microbiome of allo-HSCT recipients and found that certain bacteria, including *Staphylococcus haemolyticus* and *Ralstonia pickettii*, were associated with inferior outcomes and a higher risk of mortality [93]. Another study found that pre-transplant levels of bacteria from the genera *Kingella* and *Atopobium*, as well as post-transplant levels of *Methylobacterium*, were associated with development of oral mucositis, a common complication after conditioning therapy [94].

Other body sites could be of interest for investigation into the microbiome and allo-HSCT. Skin is commonly involved in acute GVHD, and the skin microbiome has been associated with other skin inflammatory conditions. The skin microbiome of patients with or without GVHD after allo-HSCT is currently under investigation (NCT04231500). Mucosal and epithelial surfaces can also be involved in chronic GVHD, including the mouth, eyes, genital tract and lungs, and these sites have all been found to harbor a distinct local microbiome profile that warrants further study in the allo-HSCT population.

8. Conclusions

In recent years, the microbiome has been found to be important in many aspects of human health and disease. Its potential role in improving allo-HSCT outcomes and modulating immune responses is becoming increasingly evident. We have gained insights into the influence of the microbiome on GVHD, and immune response and functions. However, successful microbiome-based strategies have yet to be fully developed for allo-HSCT. Early efforts of multi-omic analyses of the microbiome and metabolome are expanding our knowledge of the microbiome in allo-HSCT, but there is an urgent need to determine cellular and molecular mechanisms underlying this relationship in preclinical models and in patients to improve the efficacy and safety of allo-HSCT.

9. Expert opinion

Allo-HSCT is an important therapy that connects three promising areas of current clinical research: personalized cancer therapy, stem cell therapeutics and immune-modulating techniques [1]. The influence of the microbiome on disease pathophysiology, in particular on acute GVHD after allo-HSCT, has been explored by many researchers over several decades. Microbiota-derived metabolites can directly or indirectly influence allogeneic immune responses (Figure 1) [95]. The gut microbiome can vary depending on patient age in children pre-HSCT and gradually develops together with the host immune system [96]. The co-evolution between gut bacteria and their hosts can confer a mutually beneficial relationship to maintain homeostasis [97]. For example, SCFAs and tryptophan metabolites, serve as messengers to induce immune activation and modulate mucosal immunity in the host [95,98]. SCFAs can further shape the cytokine profiles and drive colonic regulatory T cells (Tregs) differentiation [60,99]. Tryptophan metabolites promote Th17 cell differentiation through Stat1 activation in the presence of AhR ligands [100]. AhR was reported to drive gut innate lymphoid (ILC)-22 development, which produce interleukin-22 as an early host defense against pathogenic infections [101,102]. A comprehensive understanding of how microbiota-derived metabolites modulate the human immune system will help guide the design of microbiota-based therapies in the allo-HSCT setting. Whether administration of live organisms is feasible and effective in controlling inflammation and regulating GVHD will require additional studies.

Alteration of microbiota-derived metabolites could imply the loss of specific bacteria and affect clinical outcomes. Recent studies have found that microbial-derived components

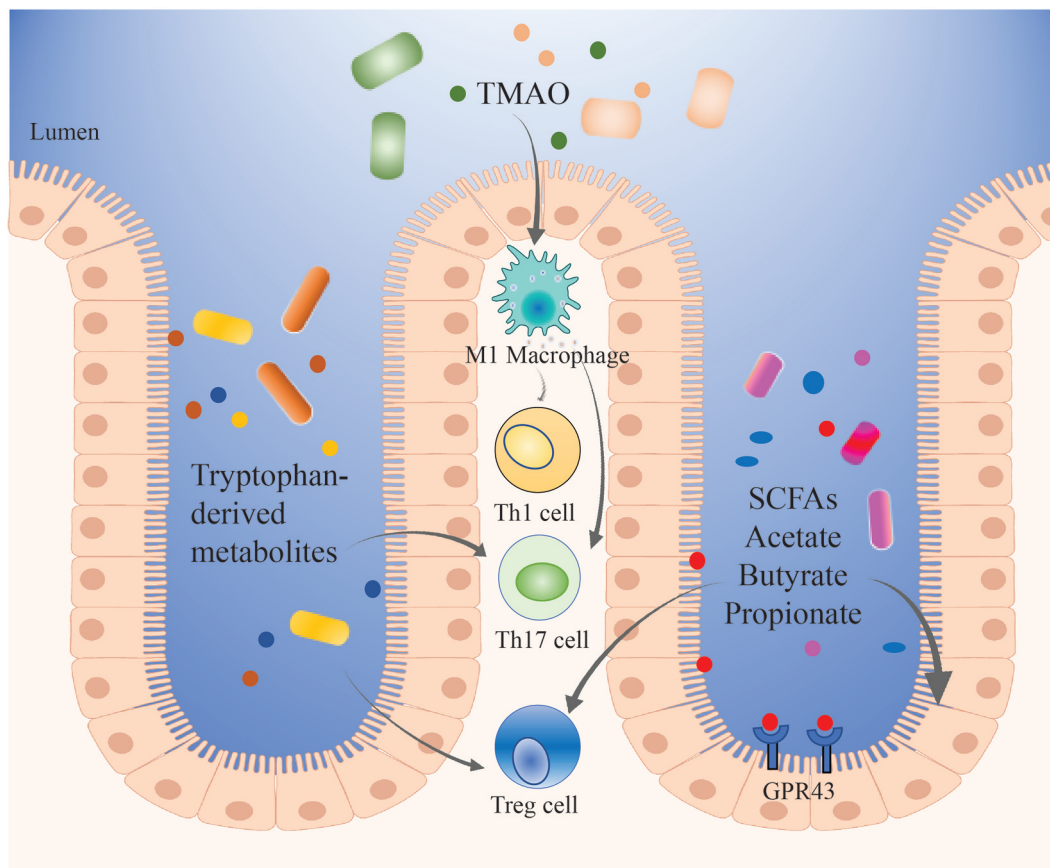


Figure 1. Microbiome-derived metabolites and their role in host modulation. Metabolic byproducts produced by commensal bacteria can impact on epithelial cell physiology and immune responses. TMAO, trimethylamine N-oxide; SCFAs, short-chain fatty acids; Treg cell, regulatory T cell.

may serve as biomarkers to predict GVHD. Plasma and urine metabolomes reflect bacterial alpha-diversity in gut [9,103], which in turn is associated with subsequent GVHD severity. 3-indoxyl sulfate, a tryptophan metabolite, was reported to be a biomarker of gut microbiota disruption [103]. Based on 16S rRNA gene deep sequencing analysis, 3-indoxyl sulfate levels are positively associated with the abundance of the Lachnospiraceae and Ruminococcaceae. In contrast, 3-indoxyl sulfate levels are negatively associated with the abundance of bacteria from the Bacilli class [10]. Specific bacterial genera present in the gut have been associated with better outcomes and improved overall survival. In particular, intestinal *Blautia* abundance near day 12 after transplantation has been significantly associated with improved outcomes, reducing the need for acute GVHD treatment and GVHD-related mortality [17].

Natural foods can supply prebiotics to help maintain the mucosal barrier. An acute change in dietary behavior, from either primarily animal to plant products, or the opposite, can rapidly alter gut microbiome within 24 hours [104]. Interestingly, an increasing pattern of *Bacteroides* and *Clostridia* can be observed in subjects placed on high protein diets [105,106]. Both prebiotics and probiotics have promise as potential interventions for allo-HSCT recipients and prospective clinical trials are underway. For example, a dietary strategy for modulating the microbiota composition could be administering dietary fiber or specific carbohydrates [75].

Recent advances in sequencing methods for microbiome profiling have paved the way in characterizing changes in gut

flora that occur in response to environmental stimuli. 16S rRNA deep sequencing is widely used to facilitate clinical translation of microbiome strategies [107,108], with sequencing capabilities being generally available and bioinformatics tools already established [109,110]. Shallow shotgun sequencing has been recommended as an alternative to 16S to provide better taxonomical resolution, while deep whole-genome sequencing (WGS) remains the gold standard [111] and additionally can provide functional data.

FMT, a more comprehensive approach to restoring an injured microbiome in comparison to probiotics, has been shown to be potentially effective in several small case series of patients with severe acute gut GVHD, and in the prophylactic setting can restore microbiome diversity via third-party FMT capsules or auto-FMT. Serious safety concerns, though, have arisen and better measures to ensure against FMT-mediated infectious complications need to be implemented. Additional studies will be needed to identify a defined minimal FMT bacterial consortium that can effectively restore the microbiome and reduce GVHD. Finally, a mechanistic understanding of the interplay between allogeneic immune reactions, microbiota, and metabolites could enable new therapeutic strategies and to provide guidance to antibiotic administration strategies.

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Declaration of interest

Robert R. Jenq has consulted for Karius, Merck, Microbiome DX, and Prolacta, and is on the scientific advisory boards of Kaleido, Maat Pharma, LIS Cure Biosciences, and Prolacta Bioscience s, and has received patent royalties licensed to Seres Biosciences. The authors have no other relevant affiliations or financial involvement with any organization or entity with a financial interest in or financial conflict with the subject matter or materials discussed in the manuscript apart from those disclosed.

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